

Manual of Rat

1 Connection

A WiFi access point has to be configured with the SSID: **LITTLELOSTROBOT** and the password: **mousewalk** The Mouse will automatically connect to that WiFi.

2 Components Assembly

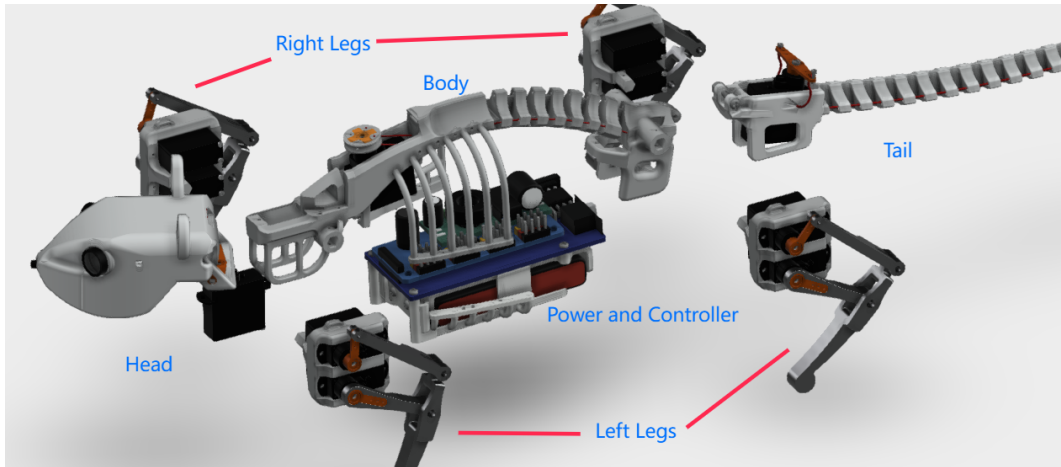


Figure 1: Exploded view of the Robot.

The modules of the rat robot can be seen in Figure 1. It is combined with four isomorphic legs, one soft actuated spine, one soft actuated tail, and a head. The spine and tail are driven by two tendons through a motor, as shown in Figure 2. The spine is driven by two tendons. For each tendon, one end is fixed on the pulley and one end is fixed in the hole in the hip. The tail is similar.

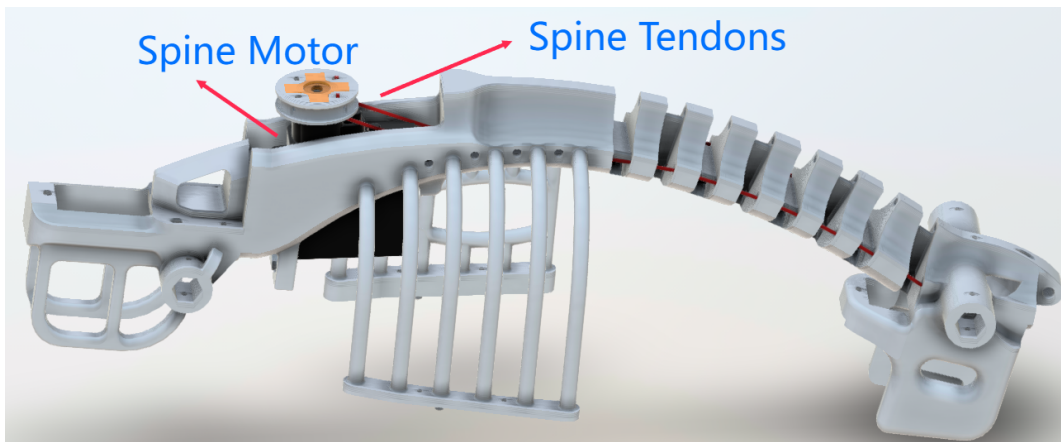
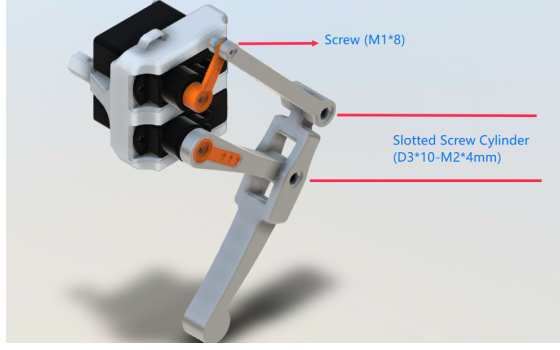
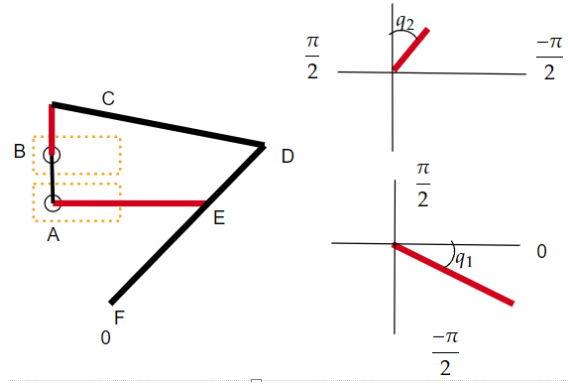


Figure 2: Installed components of the spine.



(a) Assembly of the leg modules.



(b) The initial state of the leg.

Figure 3: Caption

The four legs of the robot are isomorphic, as shown in Figure 3(a). There are several holes that need to be inserted screws. And the different parts of the leg are connected by such holes. To be attention, the initial state of the leg is important,

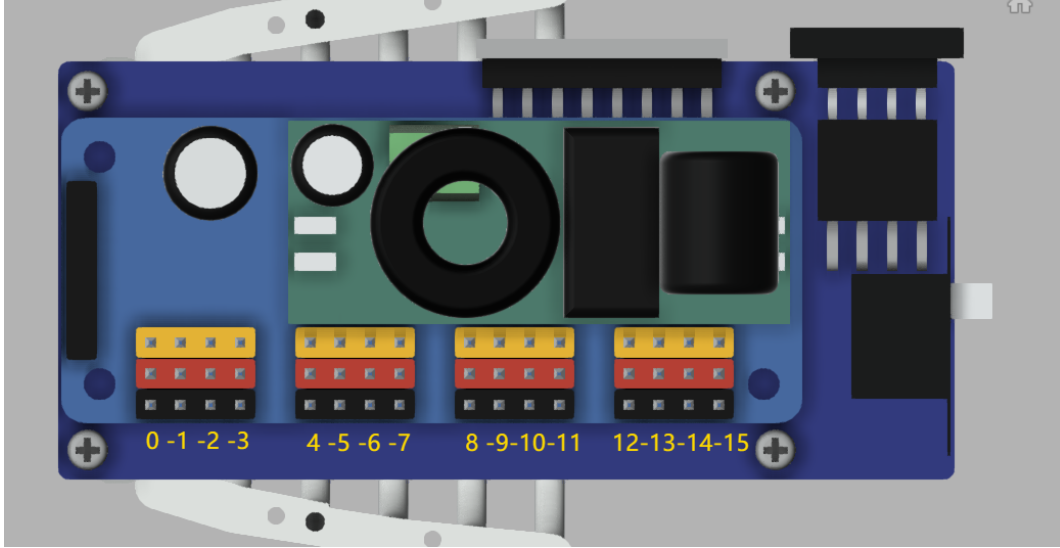


Figure 4: Control center

Pin Index	0	2	4	6	8	10	12	14
	1	3	5	7	9	11	13	15
Rat Module		Head	FL_{Leg} FL_{Coil}	FR_{Leg} FR_{Coil}	HL_{leg} HL_{Coil}	HR_{leg} HR_{Coil}	Spine Tail	

Table 1: An example table.

Finally, the connection of the servo signal wire is coded in the Arduino code. And please follow the index defined in Figure 4 and Table 1.